

Norm-boundedness in probability from convergence in distribution

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Abstract

Astra #12 rank 1. If the random Taylor data $X_n: \Omega \rightarrow \mathbf{CoeffPair}$ converge in distribution to Z , their norms are bounded in probability: $\forall \eta > 0 \exists M \forall^\infty n, \mu\{\|X_n\| > M\} \leq \eta$ (`normBounded_of_tendstoInDistribution`). The tail $\mu'\{\|Z\| \geq m\}$ tends to 0 by continuity from above, and the portmanteau theorem on the closed set $\{\|a\| \geq m\}$ bounds the limsup of $\mu\{\|X_n\| \geq m\}$ by it. Hence the tightness hypothesis of unit 152 is automatic and the ordered normalised remainders converge in distribution for any measurable Taylor data converging in distribution (`tendstoInDistribution_normA'`, `tendstoInDistribution_normB'`). Zero sorry/axiom.

1 The things formalised

```
theorem tendsto_measure_norm_ge_atTop (Z :  $\Omega' \rightarrow \mathbf{CoeffPair}$ ) (hZ : AEMeasurable Z ' ) :
  Tendsto (fun m :  $\mathbb{N} \Rightarrow ' ((fun n \Rightarrow \|Z\|) '1 \text{ Set.Ici } (m : \mathbb{N}))) \text{ atTop } (0)$ 
theorem normBounded_of_tendstoInDistribution (X :  $\Omega \rightarrow \mathbf{CoeffPair}$ ) (Z :  $\Omega' \rightarrow \mathbf{CoeffPair}$ )
  (hX : TendstoInDistribution X 1 Z (fun _ => ) ' ) :
  : ENNReal, 0 <  $\eta \rightarrow M : \mathbb{N}, 0 \leq M \quad n \text{ in } 1, \{ \|X_n\| < M\}$ 
theorem tendstoInDistribution_normA' ... (X) (hXm :  $n \mapsto \|X_n\|$  Measurable (X n)) (Z)
  (hX : TendstoInDistribution X 1 Z (fun _ => ) ' ) (Nseq) (hN : Tendsto Nseq 1 atTop)
  ( ) (h : polesBelowGen h h k k p p T) :
  TendstoInDistribution (fun n => normA ... (Nseq n) (X n )) 1
  (coeffA h h k k Z) (fun _ => ) '
theorem tendstoInDistribution_normB' ... : ... normB ... coeffB ... Z
```

2 Mathematics

Given $\eta \in (0, \infty)$ pick $m \in \mathbb{N}$ with $\mu'\{\|Z\| \geq m\} < \eta/2$; this exists because the sets $\{\|Z\| \geq m\}$ decrease to $\emptyset$ and μ' is finite. The set $F = \{a : \|a\| \geq m\}$ is closed, so the portmanteau theorem for weak convergence of the laws gives $\limsup_n \mu\{X_n \in F\} \leq \mu'\{Z \in F\} < \eta/2 < \eta$, hence eventually $\mu\{\|X_n\| > m\} \leq \mu\{X_n \in F\} < \eta$. As Astra #12 stressed, this is norm-boundedness in probability, not compact tightness (balls in ℓ^1 are not compact); it is exactly what the ball estimates of units 149–152 need.

3 Paper-facing meaning

The second conclusion of `thm:strataempiricalexpansion` now holds at chart level for $d = 2$ under the sole probabilistic hypothesis “the Taylor data converge in distribution in the weighted coefficient space”: the deterministic bound of unit 150 and the tightness hypothesis of unit 152 are removed.

The paper's separate tightness lemma (appendix prop:convergence (i)) is thus not needed as an extra input at this level; it enters only through the (unformalised) functional CLT that supplies the convergence in distribution itself.

4 Lean notes

- `tendsto_measure_iInter_atTop` wants `NullMeasurableSet` for each set (`AE measurable.nullMeasurableSet_preimage`), antitonicity, and one finite value; the empty intersection is `exists_nat_gt` after `simp`.
- `ProbabilityMeasure.limsup_measure_closed_le_of_tendsto hX.tendsto hF`, then `simp only [ProbabilityMeasure.coe_mk]` and `Measure.map_apply_of_aemeasurable` (through a universally quantified `have` fed to `simp only`) turn the pushforward measures into $\mu\{X_n \in F\}$; `eventually_lt_of_limsup_lt` closes with the default boundedness `autoParam` in `ENNReal`.
- The case $\eta = \top$ is handled separately (`ENNReal.half_pos` needs $\eta \neq 0$, `half_le_self` is unconditional); set membership `{ m < ||X n || }` must be unfolded with `show` before `le_of_lt`.